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Zoo_404

Current Mid_Term Papers

Fall 2022_23 (Jan 2_14)

Most Repeated Questions
Full Solution

Date of Collection: 08/01/2023

ZOOLOGY
LEGENDS

"Don't Stop When you Tired, Stop When you Done"

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(There is Little Possibility of Any Mistake, if you find Let us to Know)

(All Ans Are Taken From PPTs)

MCQS:

→ Hydrolysis of ATP is carried out by specific **"ATPases"** present in the membrane.

→ **Seven** hypothalamic hormones which act on anterior pituitary gland (adenohypophysis).

→ **ADH** acts by increasing water permeability of renal tubules

→ **Graded potentials/Electrical** are produced for short distance conduction at sensory and postsynaptic membranes.

→ Gonadotropes— produce gonadotropic hormones **(luteinizing hormone, LH and follicle stimulating hormone, FSH)**

→ The membrane of SR in the triad region has specialized Ca2+ ion channel proteins called **ryanodine receptors.**

→ Oxytocin & Antidiuretic hormone (ADH) or vasopressin Both hormones are peptides, containing **nine amino acid** residues.

→ In a relaxed muscle fiber, **tropomyosin** blocks crossbridging b/w myosin and actin filam

→Physiological Causes of Muscle Fatigue Ion imbalances occur due to inadequate functioning of the **Na+/K+** pump.

→Thyrotropin releasing hormone, TRH, has only **3 amino acids**

→**Catecholamines** affect contraction of smooth muscles and induce vasoconstriction.

→Transport proteins are very **specific**.

→The resting membrane is highly permeable to **K+.**

→**Ligands** are second messenger molecules and neurotransmitters.

→The function of AChE is to hydrolyze ACh into **choline and acetate.**

→ Specialized Ca2+ ion channel proteins called **ryanodine receptors** in the **membrane of SR**

→Types of feedback system___**3**

→Facilities diffusion is ____ **Passive diffusion**

→Who purposed Donna equilibrium___**1911 Frederick**

→Which is depolarization phase of action potential___**rising phase.**

→Troponin complex **40nm** of thin filaments.

→Which type of hormone excrete pineal gland__**melatonin**

→How many types of regulators___ **Three**

→Which statement is incorrect regarding homeostasis? ().

→Antidiuretic hormone, for example, can increase the water permeability of the renal collecting duct in mammals by as much as____**10 times**

→Diffusion is a passive transport___**Passive diffusion**

→The nondiffusible peptide and protein molecules have many carboxyl and other anionic groups___ **Cytosol**

→Falling phase of action potential____ **Repolarization**

→Transmission is terminated at cholinergic synapses when Acetylcholine is hydrolyzed to **choline and acetate** by AChE.

→The G proteins family consists of about **20** different proteins that are composed of three subunits, α, β and γ.

→The membrane of the cell is called___**sarcolemma**

→One of the strongest pieces of evidence in support of the sliding filament theory comes from the **length-tension** relation of a sarcomere.

→Troponin complex binds to tropomyosin about every **40 nm** along the thin actin filament.

→**Long longitudinal tubules** are involved in Ca+2 sequestering activity.

→Aa elongated protein called **Caldesmon**, which in the absence of Ca2+ , binds to actin thin filaments.

→Steroid hormones have **2** receptor cells.

→Pineal gland synthesis secrete hormone_____**Melatonin**

→The **Pancreas** is a dual gland having exocrine as well as endocrine roles.

→Cell bodies are located in the ventral horn of the **gray matter** of the spinal cord.

→T,RH Has only **3** amino acids

→Falling phase_____**Repolarization**

→Steroid hormones have **Two receptors** in a cell.

→Muscles cytoplasm is called **sarcoplasm**

→Anterior lob of pituitary gland secretes **6 hormones**

→Changes are induced experimentally in the laboratory____**Acclimation**

→And produce rising phase of action potential___**Depolarization**

→Troponin complex binds to tropomyosin about every __ along the thin actin filament. **40 nm**

→**Neuromodulators** affect Neighboring neurons and Modify their behavior at Once.

→Cell bodies are located in ventral horn of ____of spinal cord_____ **Gray matter**

→Which type of hormone excrete pineal gland_____**melatonin**

→Alzheimer's diseases, which of the disease not of animal resemble_____ **Dementia, loss of memory**

→One of the strongest pieces of evidence in support of the sliding filament theory comes from the **length-tension** relation of a sarcomere

→The G proteins family consists of about **20** different proteins that are composed of three subunits, α, β and γ.

→Adenosine releases how many trophies_____ **4**

→Adenosine triphosphate have , which helps them____**ADP**

→Fused and unfused en ma sy àik àik quiz tha shyad ya koi define

→Oraik hypothalamus kitny Jo Hy pictury gland pr kch aesa e tha dakh leina .

→-------has no diffusible peptides and proteins that have many carboxyl and other anionic groups____**Cytosol**

→Adrenal cortex produces two major types of steroid hormones_____**2**

→Steroid hormones have _____**Two receptors** in a cell.

→The opening of voltage-gated Na+ Channels in response to_____**depolarization**

Short Questions:

(These are Most Repeated Questions)

1- Detailed note on feedback system and negative feedback with example.

Definition

Mechanism of self-regulation of Biological processes.

Principle

Output or product itself regulates the process

Significance

Maintenance of homeostasis

Mechanism

Feedback controls regulate particular variables e.g., • Temperature • Salinity • pH

Negative Feedback Systems

• The most common form of regulation

• The end product works to stop or slow down the process

Example

Breakdown of sugar in the cells generates ATP. Excess ATP "feeds back" and inhibits an enzyme near the beginning of pathway. ATP production stops.

2- What are single unit muscles?

Cells small, spindle shaped & mononucleate.

• Found in the walls of Visceral organs:

• Alimentary canal • Urinary bladder • Ureters • Uterus

• Neurons synapse onto Few cells.

• Modulate rate and Strength of contraction.

• Cells coupled Electrically through gap Junctions.

• Entire muscle mass Contracts if only a few Cells are excited or Contracted.

3- Neurofibrils and its function (3)

Delicate threads running from cell body into the axon and dendrites.

Functions

• Give support and shape to the neuron.

• Transport substances from cell body to the distal cell processes.

4- Functions or Characteristics of hormones:

• Hormones coordinate long-term functions of animal tissues and organs. e.g., • growth and maintenance • sexual activity • reproductive cycles • modification of behavior.

• They perform many regulatory functions. e.g. • osmoregulation • Regulation of blood sugar levels • control of metabolic rate

• Many rhythmic activities of animals are due to hormones. e.g. • Molting • Sleep-wake up cycles • Hunger • Seasonal Reproductive activations • Migration in birds

• Maintain homeostasis

• Coordinate body's responses to stress

• Mediate responses to many environmental stimuli.

5- What is Triads?

In the skeletal muscle fiber, a T tubule is associated with two terminal cisternae of sarcoplasmic reticulum on its both sides. This arrangement of three associated tubes or sacks forms a structure that is called a triad.

Triads are located at the junction between the A and I bands of the sarcomere. Each skeletal muscle fiber has many thousands of triads which are visible in longitudinal sections of muscle fiber.

6- What is function of triad?

Function of Triads:

• Triads form anatomical basis of excitation contraction coupling, due to which a stimulus excites the muscle and causes its contraction.

• Due to the close proximity of T tubules and SR at the triads, an AP in T tubule causes the SR to release Ca2+.

7- What is myofibril?

Within each muscle fiber, there are many myofibrils running in parallel fashion. Myofibrils are 1-2 µm in diameter and extend the entire length of the cell. A myofibril is composed of myofilaments.

8- Donnan equilibrium effect

Discovered by Frederick Donnan in 1911. Given the name Donnan equilibrium.

“If diffusible solutes are separated by a membrane that is freely permeable to water and electrolytes but totally impermeable to one species of ion, the diffusible solutes become unequally distributed between the two compartments”

Donnan equilibrium is characterized by a reciprocal distribution of the anions and cations. This happens because the gradients are not only chemical but are also electrical. There must be electro-neutrality within both compartments. i.e. positive charges must equal negative charges.

9- T- tubules & it’s Role:

T tubules are thin internal extensions of the cell membrane. They are less than 0.1 µm in diameter. They innervate the cell at the level of Z disk. Make branching networks around the perimeter of each myofibril.

Role of T Tubules

- T tubule system provides the anatomic link between the surface membrane and the myofibrils deep inside the muscle fiber.
- When an action potential spreads over a muscle fiber membrane, a potential change also spreads along the T tubules to the deep interior of the muscle fiber.

- In the cell interior, T tubes are linked to SR. SR releases Ca2+ that permit myosin cross-bridges to attach and generate force for contraction.

10- Define physiology and write the name of laws used in physics and chemistry.

Physiology

Study of functions of tissues, organs and organ systems of animals. To understand the mechanisms operating in animals.

Physiology Obeys the Laws of Physics and Chemistry

- Ohm's Law
- Boyle's Law
- Ideal Gas Law
- Law of Gravity
- Kinetic & potential energy
- Inertia
- Momentum
- Velocity
- Drag

11- Cardiac muscle & features

• Found only in the heart. Striated muscles. Share many characteristics with skeletal muscle but differ in several important ways.

Features of Cardiac Muscles

• Cardiac muscle cells (myocyte) are mononucleate.

• Skeletal muscle cells are multinucleate.

• Cardiac fibers innervated only diffusely by both: • Excitatory sympathetic neurons • Inhibitory parasympathetic neurons

• Skeletal muscle fibers are individually innervated by an excitatory motor axon.

• Cardiac neural innervations are modulatory only.

• Contractions are induced by electrical activity of pacemaker.

• They don't produce discrete PSP.

• They only increase or decrease the strength of myogenic contractions.

12- How Acetylcholine acts as slow neurotransmitter?

The most familiar fast acting neurotransmitter substance is the Acetylcholine. In most instances, acetylcholine has an excitatory effect. It also acts as inhibitory transmitter in some instances e.g. at the peripheral parasympathetic nerve endings where it is involved in the inhibition of the heart by the vagus nerves.

Mode of Action of Acetylcholine

• When acetylcholine (ACh) is released into the synaptic cleft, it binds to the ligand-gated ACh specific receptors in the postsynaptic membrane.

• The binding causes Na+ and K+ ion channels to open briefly that produce an EPSP.

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Long Questions:

(These are Most Repeated Questions)

1-Write note on t- tubules system?

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Role of T Tubules

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- In the cell interior, T tubes are linked to SR. SR releases Ca2+ that permit myosin cross-bridges to attach and generate force for contraction.

2-Any five properties of hormones: (Remember only 5 to 6)

• They act only on specific target cells or tissues.

• Hormonal action depend on the type of receptor.

• A hormone comes into contact with all tissues in the body during transport through blood, however, only the cells containing specific receptors are affected by it.

• The action of a hormone depends on the nature of enzyme cascade linked to the receptor.

• It also depends on the effector molecules expressed in a tissue. For these reasons, a hormone can act on two or more different types of tissues and generate different types of responses.

• Hormones are active at very low concentration.

• The amount of hormone produced by an endocrine gland is small.

• It is further diluted in the blood and interstitial fluid.

• The available conc. for the target cell lies between 10- 8 to 10-12 M and the hormone is still effective.

3-Transmission of Signals in The Nervous System:

Methods of Signal Transmission

Neuronal signals are transmitted in two forms:

1. Graded, electronically conducted potentials

2. Action potentials

Coding of Signals During Transmission

• A signal is coded alternately in graded potentials and in all-or none APs.

• Graded potentials are produced for short distance conduction at sensory and postsynaptic membranes.

• APs are generated for long distance conduction.

• This also involves inter conversion of elect. & chem. methods at synapses.

Signal Transmission: Example

• Transmission of a signal generated by a stimulus that is received by a sensory neuron present in a receptor organ.

4-Define homeostasis and write its five Examples:

Defination:

• Walter Cannon (1929)

• The tendency of an organism to maintain relative internal stability

• To keep internal fluctuations in a narrow range.

• The ability to protect internal environment from the harms of fluctuations in external environment.

Homeostasis does not mean to keep a fixed internal environment.

Other Examples of Homeostasis

• Homeostatic regulation of water

• Homeostatic regulation of temperature

• Regulation of pH

• Regulation of glucose concentration

• Regulation of osmotic pressure

• Regulation of oxygen level

• Regulation of ion conc.

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5-Explain acclimation and acclimatization with effects and examples:

Acclimatization

• An adaptive change within an individual animal.

• Happens due to chronic exposure to new, naturally occurring environmental conditions.

Acclimatization Example

• An animal migrates to a high altitude area

• Faces low oxygen partial pressure

Effects

• Lung ventilation rate increases initially

• Ventilation drops to normal in few days

• Animal has acclimatized to the high-altitude

Acclimation

• Changes are induced experimentally in the laboratory

Acclimation Example

• Simulating high-altitude conditions in lab. • An animal physiologist places the animal in a hypobaric chamber.

Effects

• Breathing rate reacts in the same way.

• Animal becomes acclimated to the experimental conditions.

6-Resting Membrane Potential:

Resting Potential Values

• The non-excited or "resting" cell has a potential difference, Vrest across its membrane. • Values of Vrest vary in different types of cells. • Most neurons have an RMP of -70mV, when no impulse is being conducted.

Ions Responsible for RMP

• Plasma membranes are permeable to more than one type of ionic species.

• All diffusible ions affect the potential across a membrane in proportion to the permeability.

• Nondiffusible ion species have no effect on MP.

• Major ions responsible for Vrest : K+ , Na+ & Cl- .

7-Coding of signal transmission in nervous system

• A signal is coded alternately in graded potentials and in all-or none APs. Graded potentials are produced for short distance conduction at sensory and postsynaptic membranes.

• APs are generated for long distance conduction. This also involves inter conversion of elect. & chem. methods at synapses.

Signal Transmission: Example

• Transmission of a signal generated by a stimulus that is received by a sensory neuron present in a receptor organ.

8-How action potential is generated?

Definition

• Action potentials are the type of signals that neurons use to send information over long distances.

Stimulation

• When a stimulus is received at axon hillock, it causes some voltage-gated Na+ channels to open in the neuronal membrane.

• Resultantly Na+ start to diffuse in.

• This results in a local depolarization due to increase in the Vm.

• A depolarization in the range of -55 to -30 mV is the threshold that triggers an AP.

• If the intensity of stimulus is less, it causes a subthreshold depolarization that fails to generate an AP.

9-Terminal cisternae of Sarcoplasmic Reticulum? 5 marks

SR Composed of two major parts:

• Terminal cisternae

• Longitudinal tubules

• **Terminal cisternae** are larger chambers that are closely associated with T tubules (12 nm apart). • These are the primary site of calcium release as they have channel proteins that open with AP in T tubules.

• Long longitudinal tubules run between the terminal cisternae and surround the myofibrils.

• These are the locations where ion channels for calcium ion absorption (calcium pumps) are most abundant.

• So, they are involved in Ca2+ sequestering activity.

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JazakAllaH

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